Tunable 1,3-Lithium Shift of Propargylic/Allenylic Lithiums Formed by Conjugate Addition of 4-Aryl-3-butyn-1-enes with Organolithiums and the Subsequent Transmetalation/Pd(0)-Catalyzed Coupling Reaction with Aryl Halides

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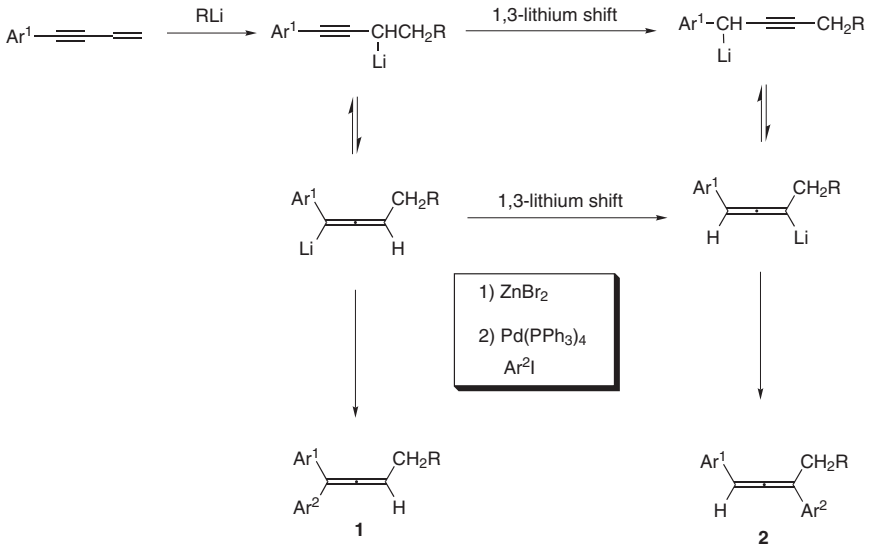
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Abstract: A protocol was established to control the 1,3-lithium shift in propargylic/allenylic species formed by the conjugate addition of 4-aryl-3-butyn-1-enes with organolithiums. Subsequent transmetalation and Pd(0)-catalyzed cross-coupling with aryl halides afforded 1,1-diarylallenes and 1,3-diarylallenes, respectively, the structural diversities of which are determined by the easily availability of both 4-aryl-3-butyn-1-enes and the organolithium reagents.

Key words: conjugated enynes, organolithiums, conjugate addition, Negishi coupling, allenes

Allenes, a class of compounds possessing two cumulative carbon-carbon double bonds, have aroused much attention due to its unique reactivity in organic synthesis.[1,2] One challenge is to make allenes with a high selectivity.[3,4] Recently, we reported a controlled 1,3-lithium shift reaction of propargylic/allenylic lithiums, which was followed by sequential transmetalation and Pd(0)-catalyzed cross-coupling with aryl halids to afford 1,1-diaryl allenes or 1,3-diaryl allenes highly selectively,[5] however, the scope of this reaction is very limited in terms of the starting alkynes. In this paper we wish to report a new protocol controlling the 1,3-lithium shift in propargylic/allenylic species formed by the conjugate addition of 4-aryl-3-butyn-1-enes with organolithiums.[6] Subsequent transmetallation and Pd(0)-catalyzed cross-coupling with aryl halides afforded 1,1-diarylallenes and 1,3-diarylallenes, respectively (Scheme 1).

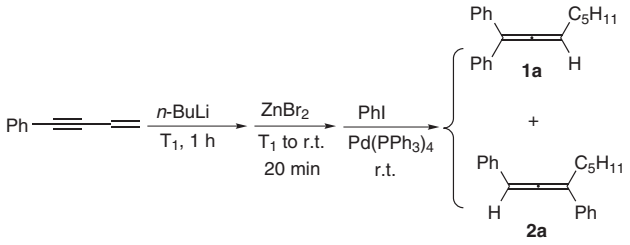
We initially tested the effect of temperature on carbolithiation, which led to the conclusion that lower carbolithiation temperature (T_1) favors the formation of 1,1-diphenylocta-1,2-diene **1a** while higher carbolithiation temperature favors the formation of 1,3-diphenylocta-1,2-diene **2a** (Scheme 2). However, although the ratios of the two products can be tuned by conducting the carbolithiation at different temperature, the yields were lower when the reactions were carried out at a higher temperature.



Scheme 1

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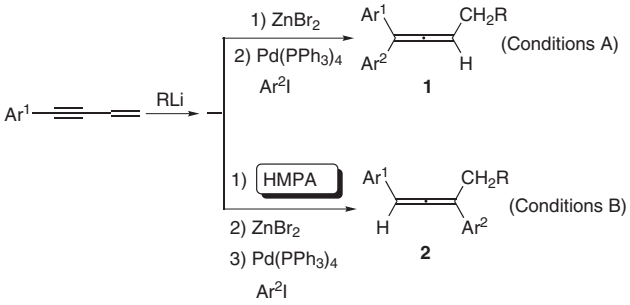
Scheme 2

Furthermore some additives were tested for the 1,3-lithium shift reaction: (1) pyridine, *n*-Bu_4NI, *N*,*N*-diisopropylethylamine and DMAP have little effect on the 1,3-lithium shift. (2) Although TMEDA has some interesting effect,[5] the reaction was not easily reproducible. (3) Through trial and error, we finally established a set of reproducible conditions to realize a clean 1,3-lithium shift. After 4-phenyl-3-butyn-1-ene was carbolithiated with *n*-BuLi at –78 °C for one hour, we added HMPA (2 equiv) at –78 °C, which was followed by warming up to room temperature naturally, subsequent transmetallation and Pd(0)-catalyzed cross-coupling led to the exclusive formation of **2a** in 83% yield, indicating a complete 1,3-lithium shift (Table 1, entry 2).

Thus, different conditions were established for the highly selective formation of 1,1-diarylallenes **1** or 1,3-diarylallenes **2**. Conditions A affording 1,1-diarylallenes **1**: 4-phenyl-3-butyn-1-ene was carbolithiated with *n*-BuLi at –78 °C for 1 hour, followed by transmetallation and coupling at room temperature. Conditions B: 4-phenyl-3-butyn-1-ene was carbolithiated with *n*-BuLi at –78 °C for 1 hour, followed by additon of HMPA (2 equiv) at –78 °C, then warming up to room temperature naturally. Subsequent transmetallation and Pd(0)-catalyzed coupling reaction afforded 1,3-lithium shift product **2**.

Some typical results are summarized in Table 1, indicating that: (1) 1° through 3° organolithium reagents could be used; (2) these two sets of reaction conditions are

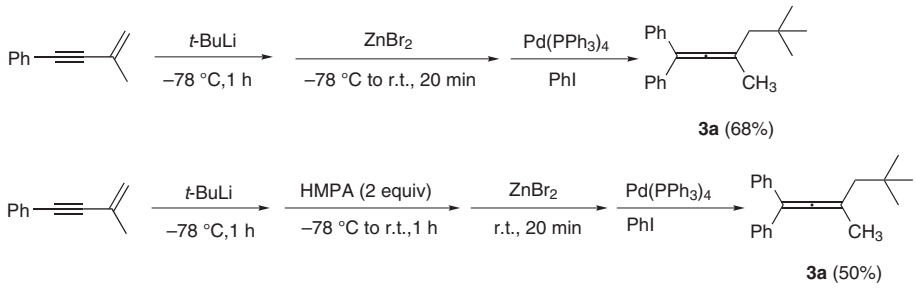
Table 1 Reactions under Conditions A and B for the Selective Synthesis of 1,1-Diphenylallenes **1** and 1,3-Diphenylallenes **2**



Entry	Ar^1	R	Ar^2	Conditions[c]	Yield (%)[a]
1	Ph	*n*-Bu	Ph	A	81 (**1a**:**2a** = 100:0)
2	Ph	*n*-Bu	Ph	B	83 (**1a**:**2a** = 0:100)
3	Ph	*t*-Bu	Ph	A	89 (**1b**:**2b** = 100:0)
4	Ph	*t*-Bu	Ph	B	88 (**1b**:**2b** = 0:100)
5	Ph	*t*-Bu	*p*-$MeOC_6H_4$	A	95 (**1c**:**2c** = 100:0)
6	Ph	*t*-Bu	*p*-$MeOC_6H_4$	B	66 (**1c**:**2c** = 0:100)[b]
7	Ph	*t*-Bu	*p*-CNC_6H_4	A	98 (**1d**:**2d** = 100:0)
8	Ph	*t*-Bu	*p*-CNC_6H_4	B	67 (**1d**:**2d** = 0:100)[b]
9	Ph	*s*-Bu	Ph	A	88 (**1e**:**2e** = 100:0)
10	Ph	*s*-Bu	Ph	B	76 (**1e**:**2e** = 0:100)
11	*p*-MeC_6H_4	*n*-Bu	Ph	A	70 (**1f**:**2f** = 100:0)
12	*p*-MeC_6H_4	*n*-Bu	Ph	B	52 (**1f**:**2f** = 0:100)

[a] The isolated yields based on aryl halides.
[b] 3 Equiv of HMPA were used.
[c] Conditions A: 4-phenyl-3-butyn-1-ene was carbolithiated with *n*-BuLi at –78 °C for 1 h, followed by transmetallation and coupling at r.t.; Conditions B: 4-phenyl-3-butyn-1-ene was carbolithiated with *n*-BuLi at –78 °C for 1 h, followed by additon of HMPA (2 equiv) at –78 °C, then warming up to r.t. naturally. Subsequent transmetallation and Pd(0)-catalyzed coupling reaction afforded 1,3-lithium shift product **2**.

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Scheme 3

general for differently substituted 4-aryl-3-butyn-1-enes and aryl halides; (3) the yields of the reactions are from good to excellent.

The reaction of 4-phenyl-2-methyl-3-butyn-1-ene used both conditions A and B afforded only 1,1-diphenylallene **3a**, indicating the importance of the proton at the 2-position of the starting material for the 1,3-lithium shift (Scheme 3).

In conclusion, we have developed a convenient method to control the 1,3-lithium shift in propargylic/allenylic species in situ formed by the conjugate addition of 4-aryl-3-butyn-1-ene with organolithium reagents in an excellent selectivity. Due to the easily availability and structural diversities of both starting compounds, this method will be very useful for the selective synthesis of allenes. Further studies on the scope and mechanism of this methodology are being carried out in our laboratory.

Acknowledgment

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